

Sutanoya Chakraborty

Research Fellow, [Indian Statistical Institute, Kolkata](#)

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Summary

I am a *research fellow* at [Indian Statistical Institute, Kolkata](#), working under the supervision of [Prof. Arijit Ghosh](#). My primary research interest lies in discrete and computational geometry.

Education

[Tata Institute of Fundamental Research, Centre for Applicable Mathematics](#) BENGALURU, INDIA

M. Sc. in Mathematics

- ☐ All India **rank 2** in CSIR-NET (National Eligibility Test) examination in Mathematics.

[Institute of Development Studies Kolkata](#)

KOLKATA, INDIA

M. Phil. in Development Studies

- ☐ Specializing in Economics.
- ☐ **Dissertation:** “*Urban Informal Workers in Print News Media - A Study of Newspapers from Kolkata*”.

[University of Kalyani](#)

KOLKATA, INDIA

M. A. in Journalism

[Bidhannagar Government College, University of Calcutta](#)

KOLKATA, INDIA

B. Sc. in Mathematics

[Bagbazar Multipurpose Girls' High School](#)

KOLKATA, INDIA

10+2 (Higher Secondary Education)

Publications 📖

- ☐ [Dimension Independent Helly Theorem for Lines and Flats](#) **Discrete and Applied Mathematics**
389, Pages 221–229, **2026**
S. Chakraborty, A. Ghosh, S. Nandi
 - ☐ [Countably Colorful Hyperplane Transversal](#) **Studia Scientiarum Mathematicarum Hungarica: Combinatorics, Geometry and Topology** **2026**
S. Chakraborty, A. Ghosh, S. Nandi
 - ☐ [Stabbing Boxes with Finitely Many Axis-Parallel Lines and Flats](#) **Discrete Mathematics**
348(2), **2025**
S. Chakraborty, A. Ghosh, S. Nandi
 - ☐ [Language Models are Crossword Solvers](#) **NAACL 2025** (main)
Pages 2074–2090
S. Saha, S. Chakraborty, S. Saha, U. Garain
 - ☐ [No Infinite \$\(p, q\)\$ -Theorem for Lines Piercing Compact Convex Sets in \$\mathbb{R}^3\$](#) *Preprint*
Under review. **2025**
S. Chakraborty, A. Ghosh
 - ☐ [A Hanani-Tutte Theorem for Cycles](#) *Preprint*
2024
S. Chakraborty, A. Ghosh
 - ☐ [Finite \$k\$ -Transversals of Infinite Families of Fat Convex Sets](#) *Preprint*
Under review. **2024**
S. Chakraborty, A. Ghosh, S. Nandi
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Experience

Visiting Researcher at University of Warsaw

Aug 2023

Visited the *Department of Mathematics, Informatics and Mechanics* at the *University of Warsaw, Poland*. Collaborated with [Prof. Kunal Dutta](#) on geometric problems.

Convex and Discrete Geometry Summer School

Aug 2023

Participated in the *Convex and Discrete Geometry Summer School* held at *Alfréd Rényi Institute of Mathematics, Budapest, Hungary*.

Teaching and Presentations

- ☐ *Instructor* at Summer Workout in Mathematics (SWIM), TIFR, Bengaluru.
- ☐ *Teaching assistant* for the Discrete Mathematics course at ISI, Kolkata.
- ☐ *Teaching assistant* for the Algorithms course at ISI, Kolkata.
- ☐ *Teaching assistant* for the Data Structures course at ISI, Kolkata.